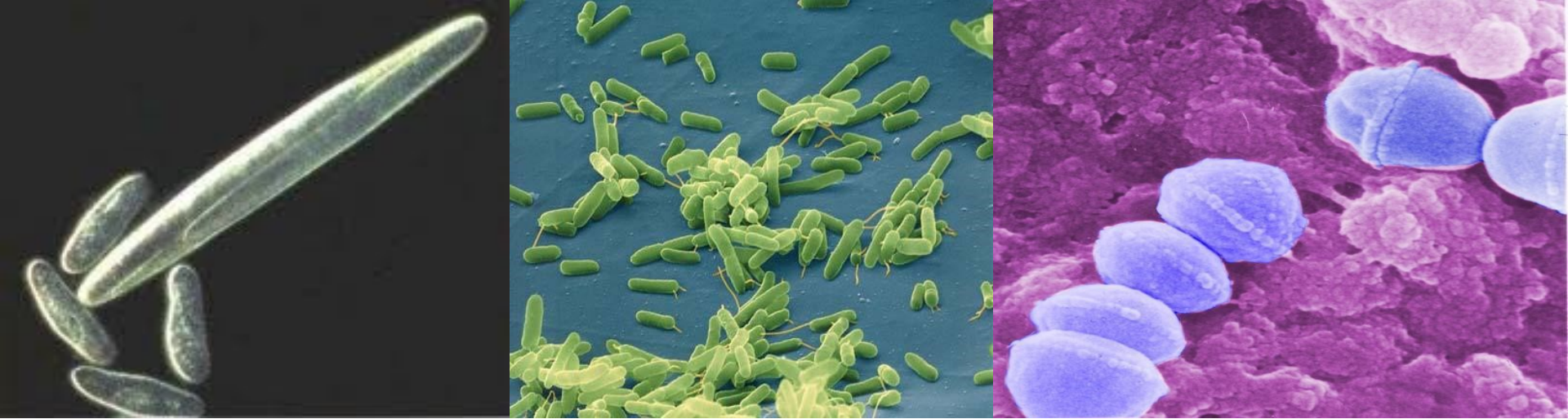




Structure and Function of a Bacterial cell

P. Neelamanie Yapa



Bacterial cell size

Vary in sizes. As small as about 0.2 μm to some cells as 700 μm the largest prokaryotes.

Typical prokaryote cells *E. coli* - 1x2 μm

Ex. *Epulopiscium fishelsoni* are huge which cells being longer than 600 μm .

This closely related to *Clostridium* and found in the gut of the surgeonfish. A unique symbiosis in the gut of a tropical herbivorous surgeonfish.

It has unusual cell division and contains multiple copies of its genome. Multiple offspring are formed and released from the mother cells of the *Epulopiscium*



Acanthurus lineatus, the striped surgeonfish



Thiomargarita namibiensis

Epulopiscium fishelsoni

Discovered in 1999 off the coast of Namibia, *Thiomargarita namibiensis* has the distinction of being the largest bacterium in the world, at a remarkable volume of 3 million times that of a normal bacterial specimen. The long string of cells, named "Sulfur pearl of Namibia" because of the reflective white globules of sulfur contained in them, was found buried in the sulfur-abundant sea floor. Each ball-shaped cell can grow up to nearly 1 millimeter in diameter--a measurement usually taken in microns. Because of some unique adaptations, *Thiomargarita namibiensis* is able to survive in a high-sulfur environment with little or no oxygen. **(Largest known bacteria)**

Smaller size of the bacterial cells in general grow faster than larger cells.

In addition given amount of nutrients will support a large population of smaller cells than larger cells.

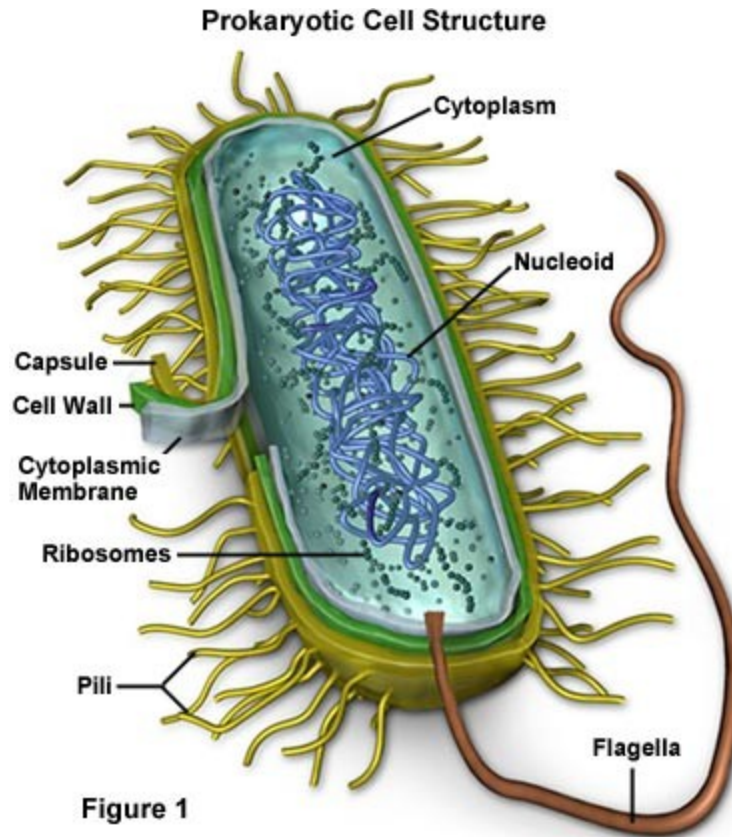
Each time cell divides chromosome replicates as well. As DNA replicates mutations occur. The more chromosome replication the greater the total number of mutations . Hence important in evolution because bacteria are also haploid. Prokaryotes are adapted well in the extreme environmental conditions and exploit more habitats.

There is a lower limit of cell size. Smallest bacteria microbiologists observed were *nanobacteria* -0.1 μm in diameter

Smaller than prokaryotic cells. Cell should have to be minimum 0.15 μm in diameter for having essential biomolecules.

Therefore the question of *nanobacteria* really living is unanswered.

Prokaryotic cell structure



Prokaryote Anatomy

Composition of a bacterial cell

Important points:

- Water makes up ~ 70%

- Macromolecules make up ~ 26% (~90% of dry wt)

- The most abundant molecules other than water are proteins (~50% of dry wt)

- Small molecules & ions make up 4% (~10% of dry wt)

How big are bacteria?

Smallest bacteria are **Mycoplasmas**, as small as 0.2 micrometers (almost as small as largest poxviruses!)

Mycoplasmas = group of bacteria that evolved to lack wall. Typically found in environments with high osmotic strength

Structure and functions

Capsule or Glycocalyx

Some have third protective covering – made up of polysaccharides. **glycocalyx** (also called **slime layer**, **capsule**). Not found in all bacteria.

Varies in thickness, rigidity.

Some suggestion that many bacteria lose layer when cultured in laboratory.

May be much more prominent in nature than thought.

Functions:

Keep the bacterium from drying out.

Protect it from phagocytosis

Facilitate attachment of the bacterium to its target cells

Capsule is the major virulence factor in the disease causing bacteria such *Escherichia coli* and *Streptococcus pneumoniae*. Noncapsulated mutants of these are avirulent

Bacterial adhesion promotes formation of biofilms, masses of bacteria encased in large aggregates of extracellular matrix. Biofilms are not well-understood, but incredibly important. Most bacteria may live largely in biofilms rather than as free organisms (the dispersal stage). Biofilms are harder to get rid of, more resistant to antibiotics.

Cell wall

Mainly composed of peptidoglycan-a protein sugar

Rigid layer, preserves shape when rest of cell is digested.

Made of peptidoglycan = polymer of **peptides** (typically 4 amino acids long, cross-linked to other chains) and **glycans** (made of alternating amino sugars)

Sugars found in peptidoglycan

N-acetylglucosamine (NAG).

N-acetylmuramic acid (NAM).

NAG-NAM sugars are linked by β -1,4 linkage).

Note: glycan chain is similar to chitin (= lobster shell), a polymer of NAG.

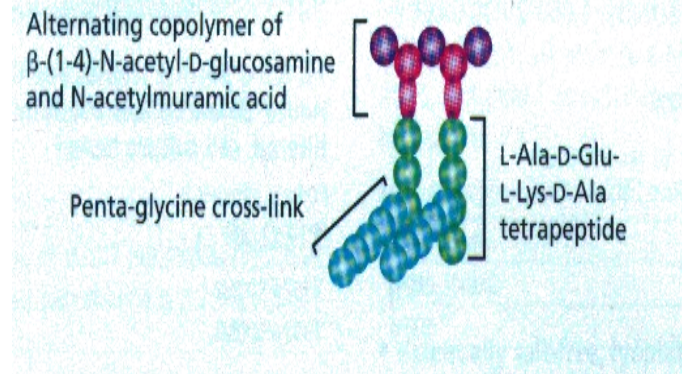
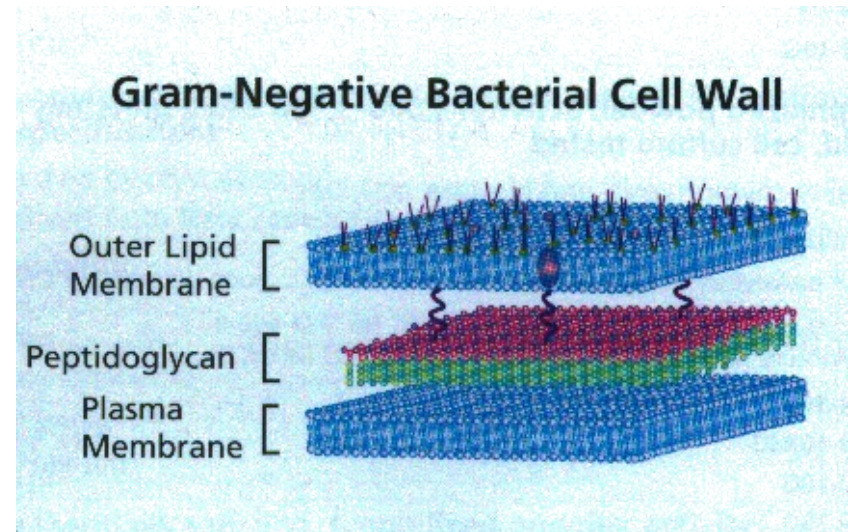
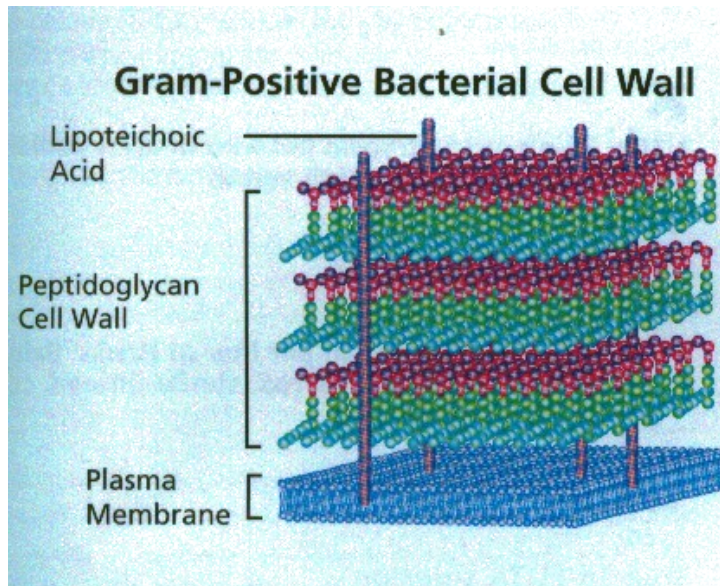
Peptidoglycan is a polysaccharide composed of two sugar derivatives

N-acetylglucosamine (NAG).

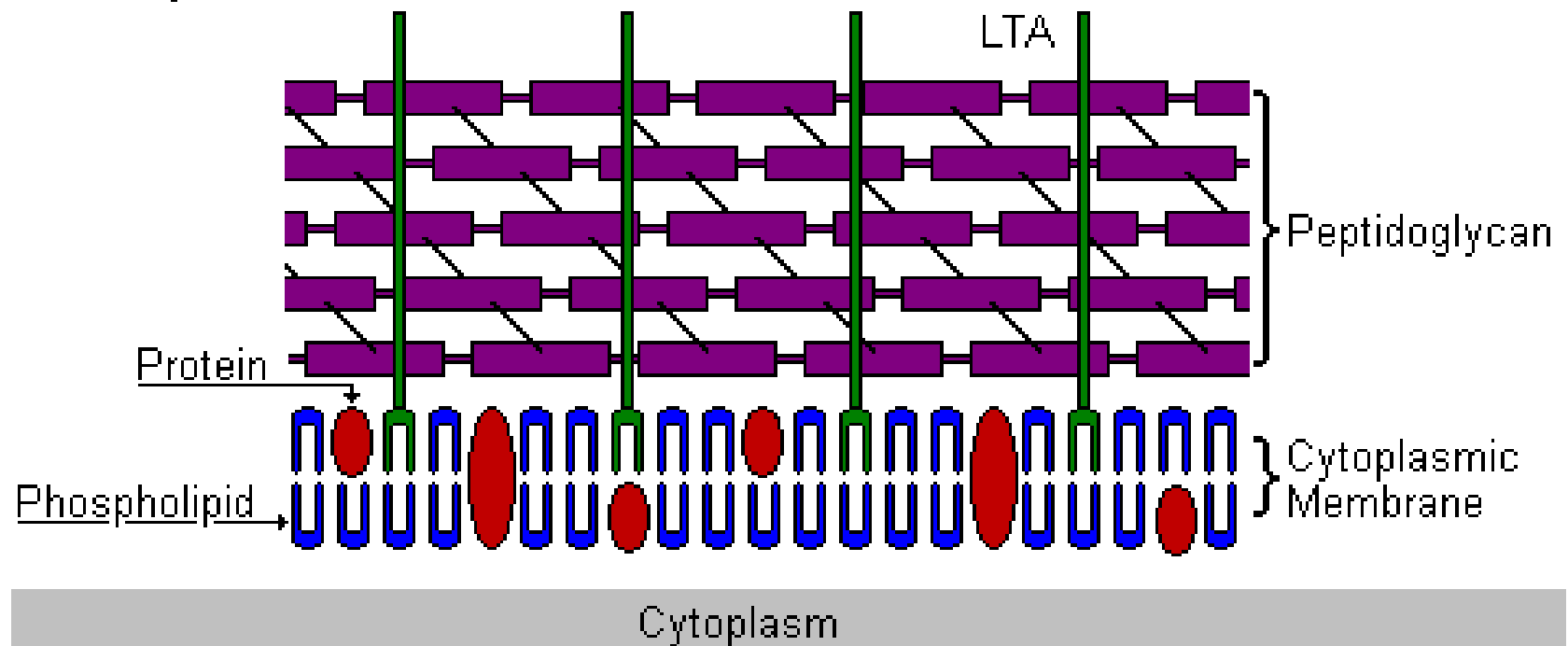
N-acetylmuramic acid (NAM). And few amino acids include L- alanine , D- alanine, D- glutamic acid and either lysine or diaminopimelic acid (DAP)

These constituents are connected to form repeating structure. – the glycan tetrapeptide

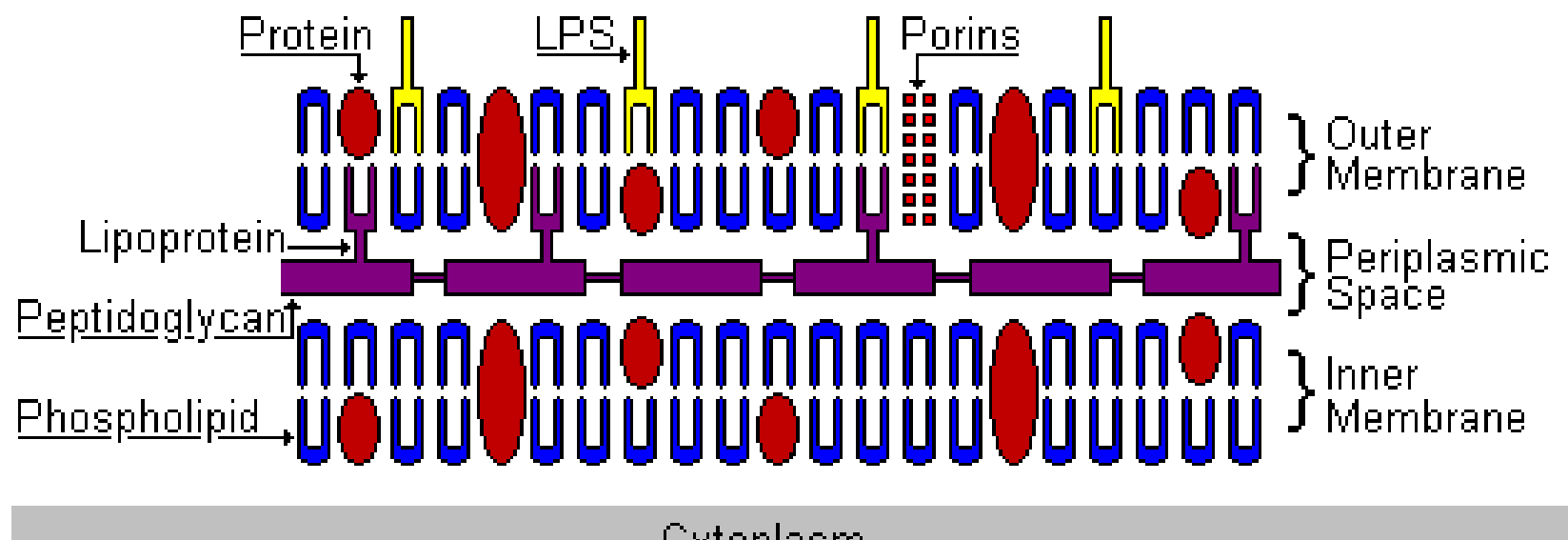
Cell wall



Gram-positive Cell Wall



Gram-negative Cell Wall



Cell wall composition varies among bacteria-
important in species differentiation.

Hans Christian Gram in 1884 found
differential staining technique

Gram positive- high percentage of
peptidoglycan arranged in several layers,
bridging of **teichoic acid**(amino
acid)links-**murein**

Gram negative-consist of thin peptidoglycan
layer but has a lipopolysaccharide outer
membrane(lipid with sugar polymers) -
toxic to a host, contribute to
pathogenecity - endotoxin

Teichoic Acid - Teichoic acids are also substituted polysaccharides containing ribitol or glycerol residues joined through diphosphoester linkages . These are generally found in the Gram-positive bacteria in association with the peptidoglycan. They constitute a major surface antigen.

Teichoic acids are found in the cell walls (wall teichoic acids), in the membranes (membrane teichoic acids)

Two types of teichoic acids known are glycerol and ribitol teichoic acids

- **Lipoteichoic acids:** Lipoteichoic acids (LTA) are found only in Gram-positive bacteria. These polysaccharides extend through the entire peptidoglycan layer and appear on the cell surface.
- **Lipopolysaccharides:** Lipopolysaccharides (LPS) are found only in Gram-negative bacteria. These structures are composed of lipid A, which binds the LPS in the outer membrane and is itself the endotoxic portion of the molecule. The polysaccharide moiety appears on the cell surface, serving as an antigenic determinant ("O antigen").

Gram negative cells have much thinner peptidoglycan, but also have **outer membrane** (OM) made of lipid, protein, and LPS (**lipopolysaccharide**). OM is porous because of **porin proteins** that create pores, allow small molecules to pass freely.

- **Periplasmic space** = space between Inner membrane and OM. May occupy as much as 30% of cell volume. Not just an entryway: contains many specialized proteins, often involved in nutrient acquisition. Ex: phosphatases, enzymes that cleave organic molecules containing phosphate, make it available for cell nutrition.

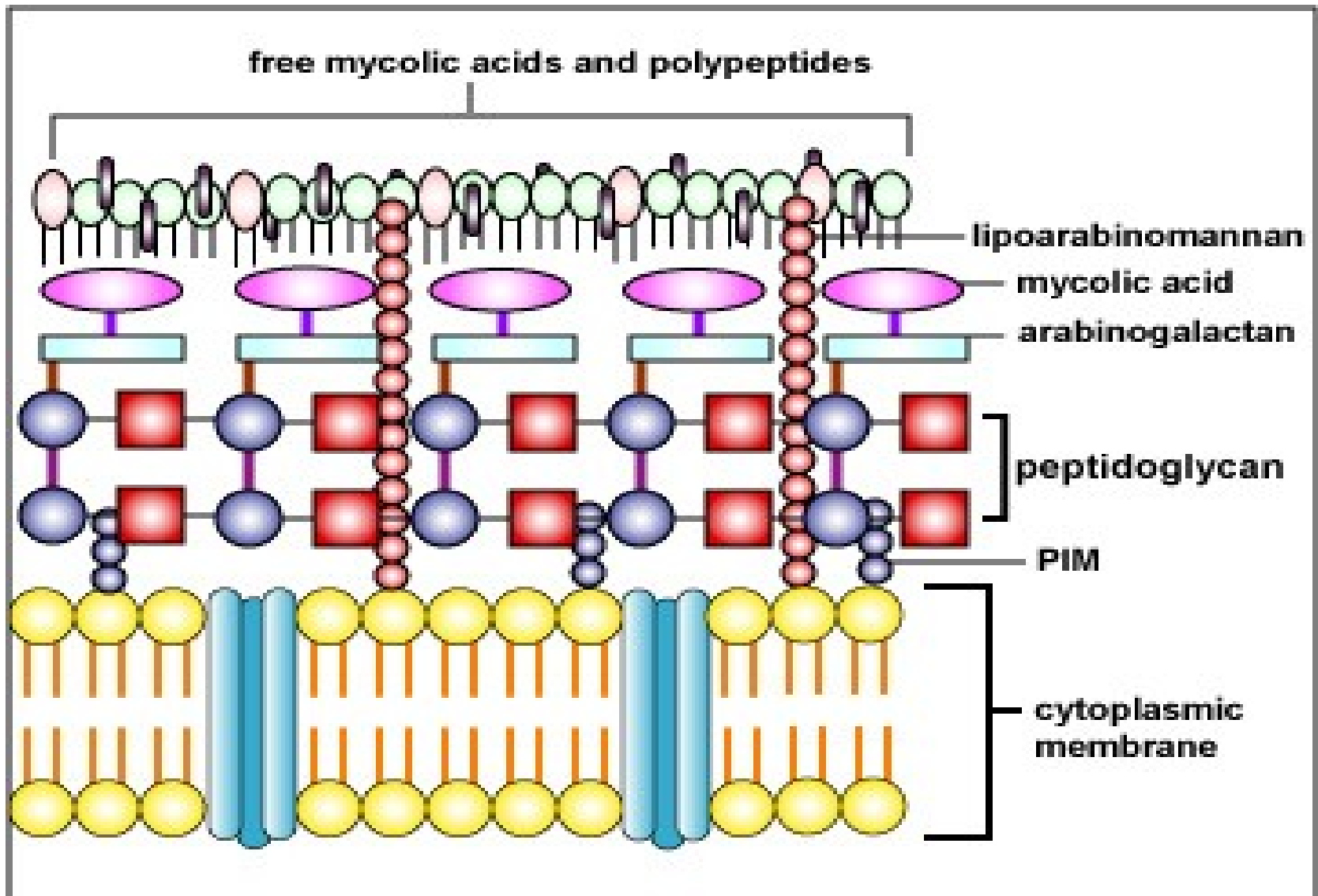
When Gram staining thicker walls of Gram + cells are dehydrated with decolourizer- cause pores diminish in size

In contrast, Gram negative walls are not strong, outer membrane destroyed and walls are too thin to retain primary stain

Gram positive – can withstand disinfectants better, so require higher concentration or longer application.

Certain antibiotics work better on one or the other type, eg. Penicillin is specific to gram positive.

Acid fast cell wall



Structure and Composition of the Acid-Fast Cell Wall

Acid-fast bacteria are gram-positive, but in addition to **peptidoglycan**, the outer membrane or envelope of the acid-fast cell wall of contains **large amounts of glycolipids, especially mycolic acids** that in the genus *Mycobacterium*, make up approximately 60% of the acid-fast cell wall.

1. The acid-fast cell wall of *Mycobacterium* has a thin, inner layer of **peptidoglycan**
2. The peptidoglycan layer is, in turn, linked to **arabinogalactan** (D-arabinose and D-galactose)

3. The arabinogalactan is then linked to an outer membrane containing high-molecular weight mycolic acids. The arabinogalactan/mycolic acid layer is overlaid with a layer of polypeptides and mycolic acids consisting of free lipids.

Like the outer membrane of the gram-negative cell wall, porins are required to transport small hydrophilic molecules through the outer membrane of the acid-fast cell wall

4. The outer surface of the acid-fast cell wall is studded with surface proteins that differ with the strain and species of the bacterium.

5. The periplasm is the gelatinous material between the peptidoglycan and the cytoplasmic membrane.

Functions of the Acid-Fast Cell Wall Components

1. The peptidoglycan prevents osmotic lysis
2. The mycolic acids and other glycolipids also impede the entry of chemicals causing the organisms to grow slowly and be more resistant to chemical agents and lysosomal components of phagocytes than most bacteria

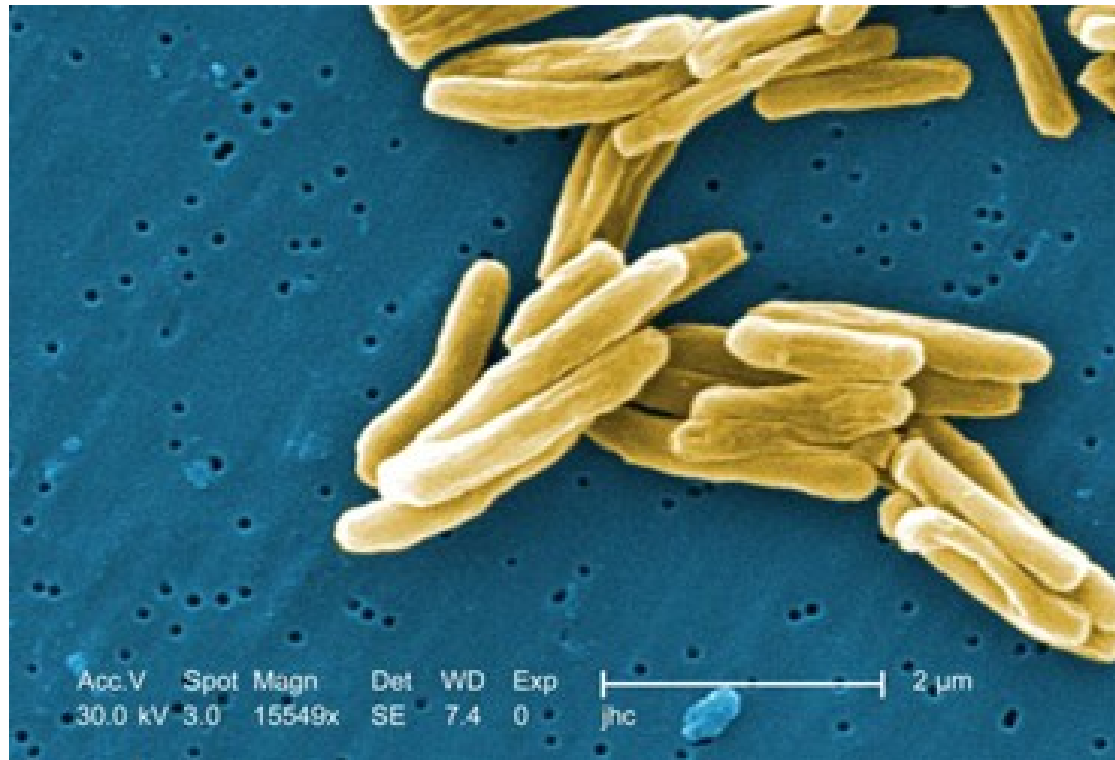
There are far fewer porins in the acid-fast cell wall compared to the gram-negative cell wall and the pores are much longer. This is thought to contribute significantly to the lower permeability of acid-fast bacteria.

3. The surface proteins in the acid-fast cell wall, depending on the strain and species, carry out a variety of activities, including:

Functioning as enzymes.

Serving as adhesins:enable the bacterium to adhere intimately to host cells and other surfaces in order to colonize and resist flushing.

4. The periplasm contains enzymes for nutrient breakdown.



Scanning Electron Micrograph of *Mycobacterium tuberculosis*

**Image provided by Dr. Ray Butler and Janice Carr.
Courtesy of the Centers for Disease Control and
Prevention**

Mycobacterium tuberculosis is much more resistant to antibiotics and disinfectants than most other bacteria. It also grows much more slowly. Why might this be?

Acid-fast bacteria such as the mycobacteria have a cell wall with a relatively impermeable outer membrane containing a waxy lipid called mycolic acid. This mycolic acid outer membrane impedes the entry of many harmful disinfectants and antibiotics but also slows the entry of nutrients needed for bacterial growth.

Osmotic effects

Water moves freely across membranes (= osmosis) to try to equalize concentration. Interior of cell = ~ 0.9% salt.

If surrounding fluid = 0.9% salt, **Isotonic**. Equal water flow in & out

If surrounding fluid < 0.9% salt, **Hypotonic**. Water flows in, presses against wall. No problem if wall is intact (can withstand 20 atmospheres of pressure

If surrounding fluid > 0.9% salt, **Hypertonic**. Water flows out; cell shrinks --> **plasmolysis**. Stops metabolism.

Effects of lysozyme and penicillin

Growing cell must break peptidoglycan at certain spots, insert new material, seal back together to grow new wall

Cell wall contains bacterial enzymes called **autolysins** to make these breaks, but carefully controlled. Mechanism not well understood.

Lysozyme = enzyme that attacks PG, breaks glycan chain. Found in animal secretions, e.g. tears, saliva, mucous membranes. More active against Gram⁺ than Gram negative cells.

Penicillin = fungi produce antibiotic, blocks formation of new peptide links in **growing** cells walls (= **transpeptidation**).

Note particular advantage of penicillin: no harmful effects on human cells, since PG is unique to bacteria. Penicillin G (the natural isolate) is less effective against many gram-negative cells, since it has difficulty penetrating the outer membrane.

Archaea: different cell walls

Cell walls in certain bacteria made of other materials, not peptidoglycan. Can be protein, glycoprotein, or polysaccharide. Some archaea have pseudopeptidoglycan, similar architecture, but N-acetyltalosaminuronic acid instead of N-acetylmuramic. All archaeal walls are resistant to lysozyme.

Cytoplasmic/cell membrane

Layer of phospholipids and proteins

Encloses the interior of the bacterium, regulating the flow of the material in and out

Since there are no other membrane bound organelles it provides a site for protein attachment

Enzyme activity

And provide surface for important metabolic pathways-eg. respiration

function: permeability barrier, prevents cell contents from leaking away. Very impermeable to polar & charged molecules (most biomolecules, also small ions such as K^+ , H^+). Only water, lipid-soluble materials pass freely.

But cells have many specific protein carriers in membrane, carry out selective transport.

Chemical Composition of membranes: ~50% lipid, 50% protein

Membrane lipids are typically phospholipids, contain two **fatty acids** joined to **glycerol** by **ester bonds** Phospholipids assemble into lipid bilayer.

Cytoplasm

where the functions of growth, replication, and metabolism are carried out.

Consist of water, enzymes, nutrients, wastes, and gases.
Also cell structures- ribosomes, a chromosome, and plasmids

DNA-single stranded circular chromosome localized in cytoplasm-nucleoid

DNA consist of code of structure of proteins

Extra chromosomal circular piece of DNA known as **plasmid** can be found in some bacteria.

Unlike the chromosome plasmids are not involved in reproduction

Plasmids replicate independently. Copies passed into daughter cells during binary fission or conjugation

They transmit special properties, such as antibiotic drug resistance, resistance to heavy metals, virulence factors for infection etc.

The ability to insert specific genes into plasmids made them extremely useful in the field of genetic engineering

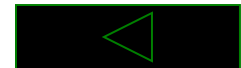
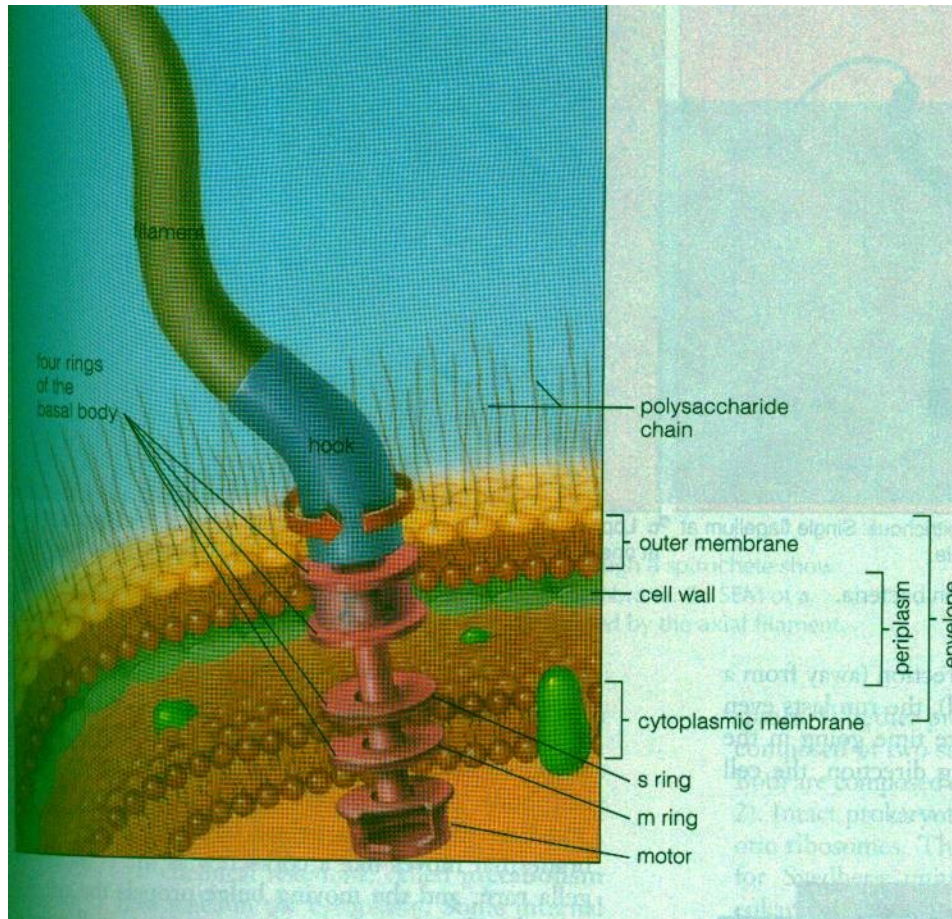
Flagella

curved filament made of **flagellin** protein: travels through hollow tube, assembles at external end.

can be arranged in two ways:

Polar flagellation: flagella attached at one (monopolar) or both (bipolar) ends.

Peritrichous flagellation: flagella attached at many sites around cell periphery. Ex: *E. coli*
attached to cell via **basal region** Provide a means of locomotion-propeller-like motion



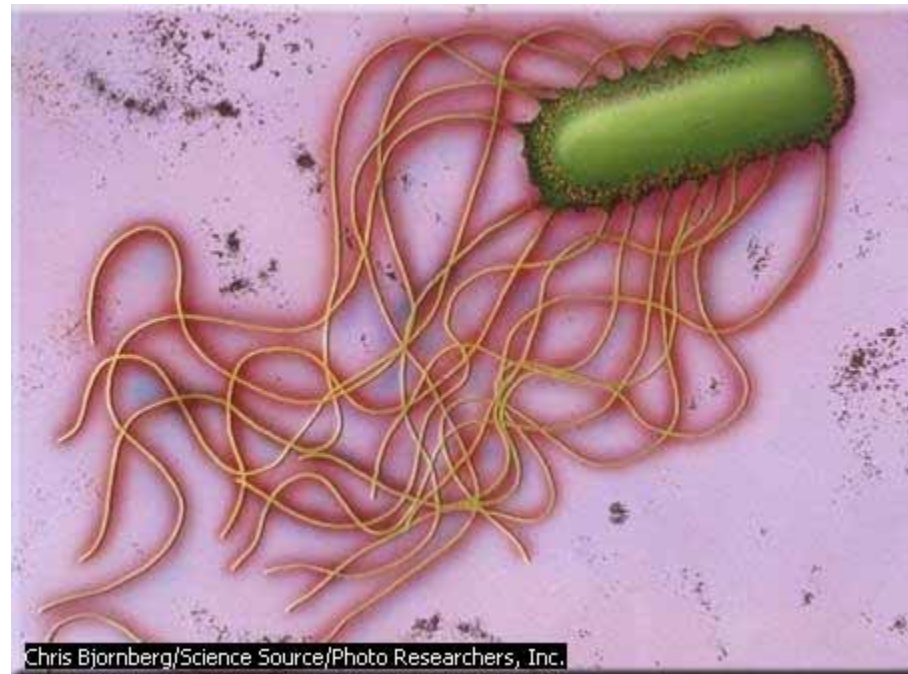
Move away from toxic chemicals, toward nutrients, photosynthetic cyanobacteria towards light

Monotrichous- one flagellum

Amphitrichous-2 flagella at both ends

Lophotrichous-tuft at one end

Peritrichous-many flagella



Chris Bjornberg/Science Source/Photo Researchers, Inc.

Prokaryotes exhibit a variety of types of tactic behaviour-ability to move in response to environmental stimuli

Chemotaxis-move towards or away from certain chemical compounds

Phototaxis-light stimuli

Magnetotaxis- magnetic stimuli

Pili or fimbriae

Small hair like projections emerging from the outside of the cell surface

Assist bacteria to attach other cells and surfaces

Without them most disease causing bacteria unable to infect

Neisseria gonorrhoeae need to attach by fimbriae

Helps to conjugation –called as sex or F pili

Ribosomes

Translate the genetic code of nucleic acid to amino acid

Smaller and slightly different composition and structure from eukaryotes

Free standing and distribute in the cytoplasm

Some antibiotics will inhibit the function of bacterial ribosomes but not eukaryote they are infecting.

Endospore/ spore of bacteria

Produced by vegetative cell that can dormant for many years and non- metabolizing until conditions are right

Non-reproductive structure used for survival-produces only one cell when spore germination.

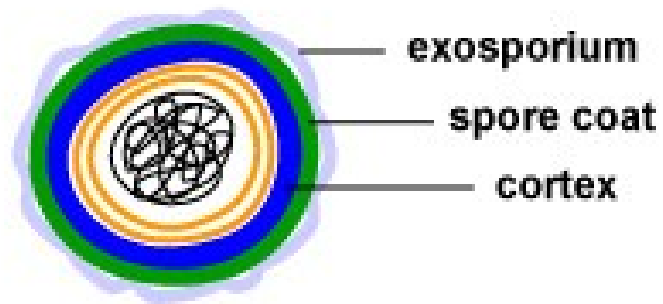
The position of endospore within a vegetative cell – central or terminal

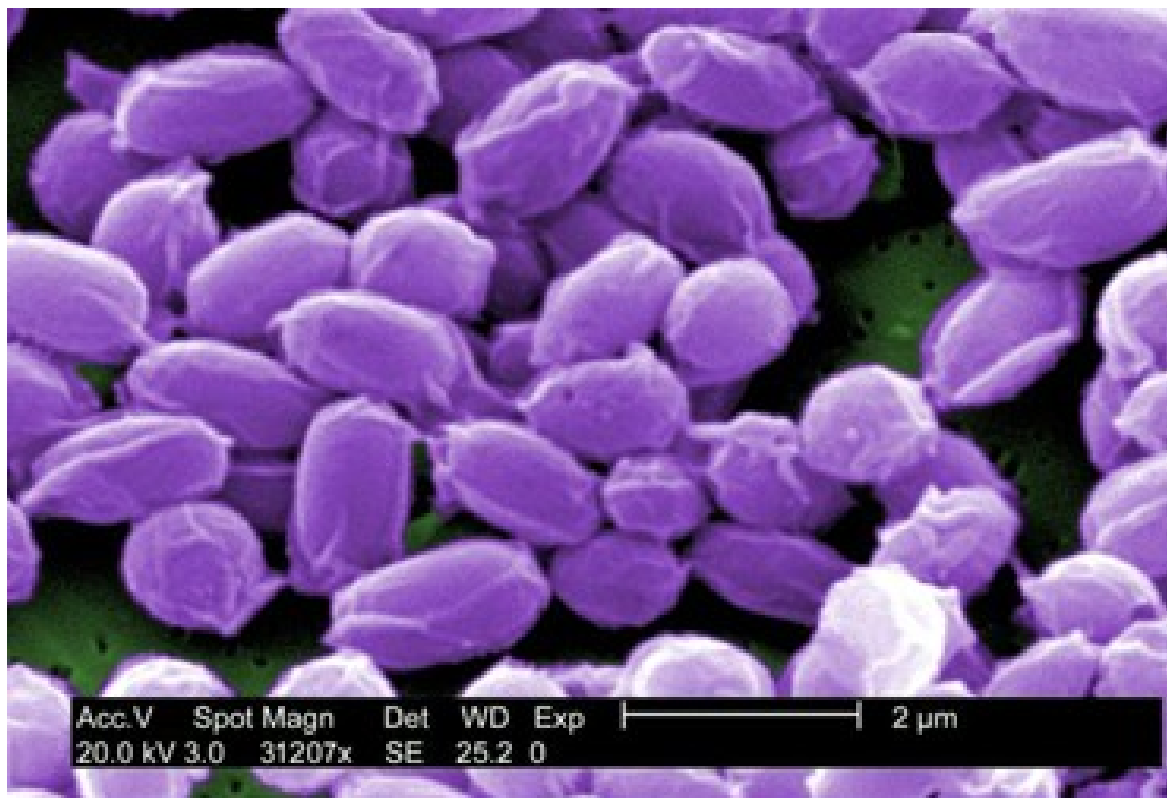
Produced only 2 genus's

Bacillus (eg. *Bacillus anthracis*, *Bacillus macerans*)

Clostridium (eg. *Clostridium tetani*, *Clostridium botulinum*)

endospore





**Scanning Electron Micrograph of the Endospores of
*Bacillus anthracis***

Bacillus anthracis is the bacterium that causes anthrax. It is usually transmitted by endospores.

Endospore Sporulation

Endospores are made through a process known as sporulation in response to extreme environmental conditions. Extreme environments include high temperatures, drying out (dessication), extremes in pH, bacteriocidal chemicals and lack of food.

When environmental conditions are favorable, the endospore germinates. Upon germination, the cell returns to its normal metabolic state, capable of reproduction.

Sporulation

First the DNA replicates and a cytoplasmic membrane septum forms at one end of the cell. A second layer of cytoplasmic membrane then forms around one of the DNA molecules (the one that will become part of the endospore) to form a forespore. Both of these membrane layers then synthesize peptidoglycan in the space between them to form the first protective coat, the cortex.

Calcium dipicolinate is also incorporated into the forming endospore. A spore coat composed of a keratin-like protein then forms around the cortex. Sometimes an outer membrane composed of lipid and protein and called an exosporium is also seen.

Finally, the remainder of the bacterium is degraded and the endospore released. Sporulation generally takes around 15 hours.

An endospore is not a reproductive structure but rather a resistant, dormant survival form of the organism.

Endospores are quite resistant to high temperatures (including boiling), most disinfectants, low energy radiation, drying, etc.

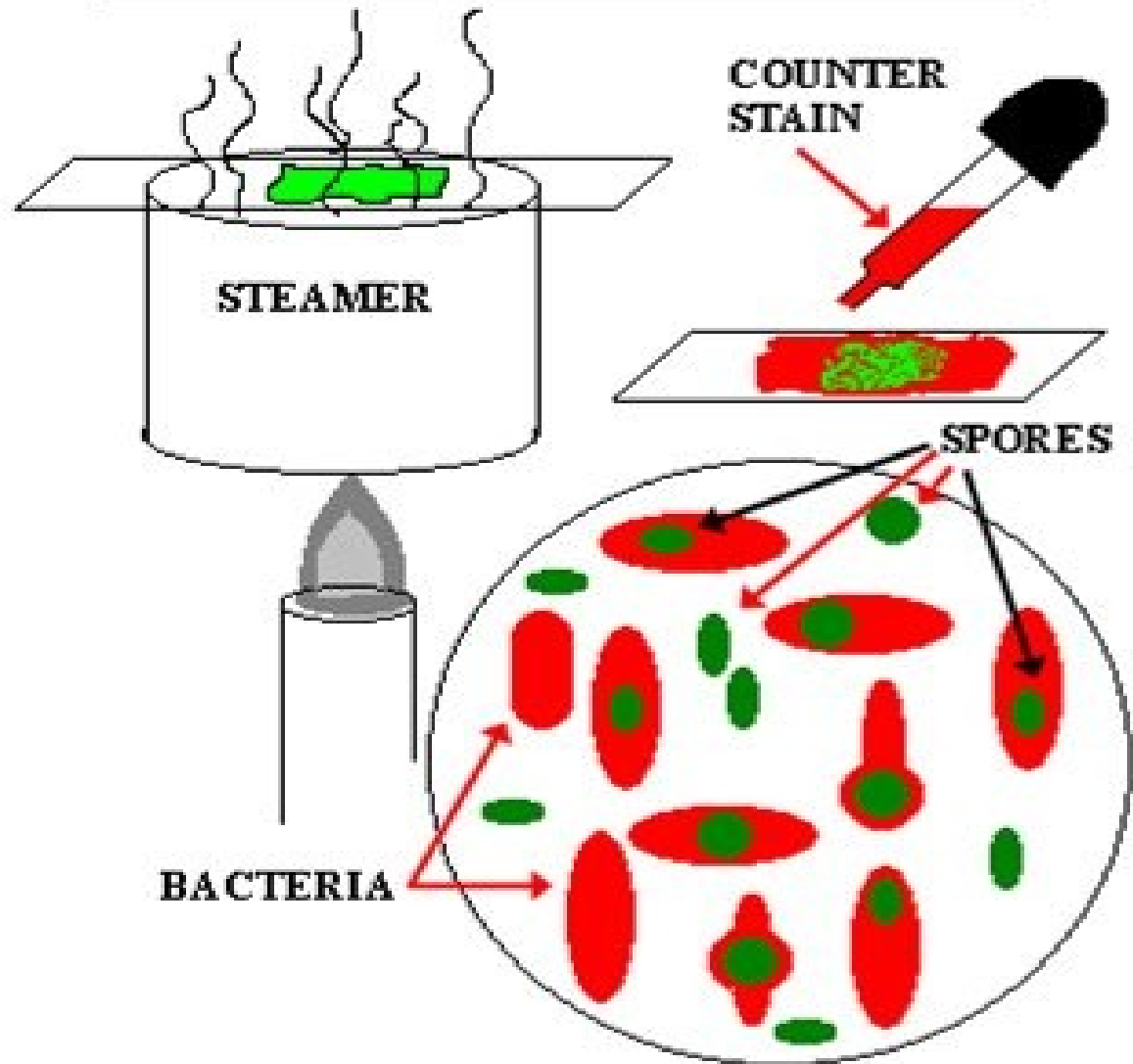
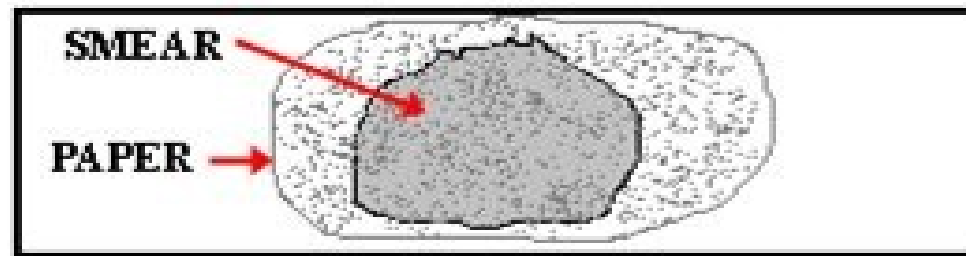
The endospore can survive possibly thousands of years until a variety of environmental stimuli trigger germination allowing outgrowth of a single vegetative bacterium .

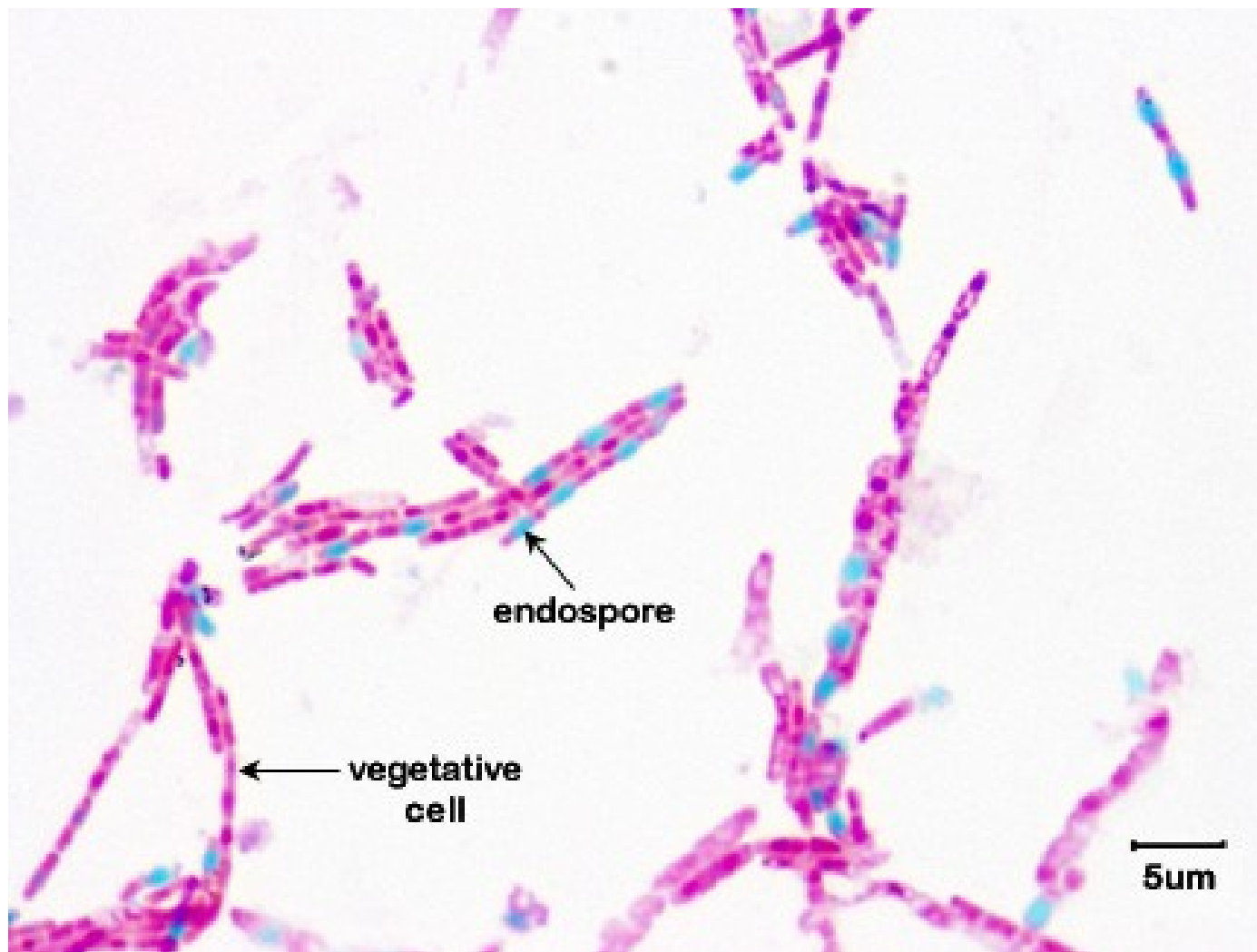
Staining Endospores

Normal staining techniques will not stain the resistant endospores. Here are the steps that are required:

- *Malachite Green*: The stain, malachite green, is forced into the spore with heat much like the carbol fuchsin was forced through the waxy mycolic acid layer of *Mycobacterium*. After flooding the slide with malachite green, it is suspended over a boiling water bath for 5 minutes and then rinsed with water.
- *Safrnin*: Once the endospore is stained, the counter stain, Safranin, provides color for the vegetative (i.e. metabolically active) cells.

—





Endospore stain of *Bacillus megaterium*

Note green endospores within pink bacilli.

Organelles Used in Bacterial Photosynthesis

There are three major groups of photosynthetic bacteria: cyanobacteria, purple bacteria, and green bacteria.

The cyanobacteria carry out oxygenic photosynthesis, that is, they use water as an electron donor and generate oxygen during photosynthesis. The photosynthetic system is located in an extensive thylakoid membrane system that is lined with particles called phycobilisomes.

The green bacteria carry out anoxygenic photosynthesis. They use reduced molecules such as H_2 , H_2S , S , and organic molecules as an electron source and generate NADH and NADPH. The photosynthetic system is located in ellipoidal vesicles called chlorosomes that are independent of the cytoplasmic membrane.

The purple bacteria carry out anoxygenic photosynthesis. They use reduced molecules such as H_2 , H_2S , S , and organic molecules as an electron source and generate NADH and NADPH. The photosynthetic system is located in spherical or lamellar membrane systems that are continuous with the cytoplasmic membrane

Other Inclusion Bodies

Cyanobacteria contain large inclusion bodies called **cyanophycin granules** that store nitrogen for the bacteria.

Cyanobacteria, thiobacilli, and nitrifying bacteria, organisms that reduce CO_2 in order to produce carbohydrates, possess **carboxysomes** containing an enzyme used for CO_2 fixation.

Purple and green photosynthetic bacteria as well as some other aquatic bacteria contain **gas vacuoles**. These are aggregates of hollow protein cylinders called gas vesicles that are permeable to atmospheric gas.

Some bacteria produce inorganic inclusion bodies in their cytoplasm, including **volutin granules** that store phosphate, and **sulfur granules** that store sulfur.

Some bacteria produce organic inclusion bodies containing either **polyhydroxybutyrate granules** or **glycogen granules** as an energy reserve.

Some motile aquatic bacteria are able to orient themselves by responding to the magnetic fields of the earth because they possess **magnetosomes**, membrane-bound crystals of magnetite or other iron-containing substances that function as tiny magnets.